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Problem Set 9

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Problem 1: Double Counting

We must prove that for all natural numbers $n \leq a \leq b \leq c$:

$$\sum_{i,j,k} \binom{a}{i}\binom{b}{j}\binom{c}{k} = \binom{a+b+c}{n}$$

where the sum ranges over all $i, j, k \in \mathbb{N}$ satisfying $i + j + k = n$.

Combinatorial Proof: Suppose we have a set of $a + b + c$ distinct items. Let this set be partitioned into three disjoint subsets: Group A of size a, Group B of size b, and Group C of size c.

We want to count the total number of ways to choose exactly n items from this entire set of $a + b + c$ items.

Method 1 (Right-Hand Side): By definition, the number of ways to choose n items from a set of $a + b + c$ items is simply $\binom{a+b+c}{n}$.

Method 2 (Left-Hand Side): Alternatively, we can categorize our choices based on how many items we take from each of the three groups. Let i be the number of items chosen from Group A, j be the number of items chosen from Group B, and k be the number of items chosen from Group C. To choose exactly n items in total, it must be true that $i + j + k = n$.

For any fixed valid triplet (i, j, k):

- We can choose i items from Group A in $\binom{a}{i}$ ways.
- We can choose j items from Group B in $\binom{b}{j}$ ways.
- We can choose k items from Group C in $\binom{c}{k}$ ways.

By the Rule of Product, there are exactly $\binom{a}{i}\binom{b}{j}\binom{c}{k}$ ways to select the n items such that the group breakdown is exactly i, j, and k.

To find the total number of ways to choose the n items across all possible valid distributions, we sum over all combinations of i, j, and k that sum to n. This gives the expression:

$$\sum_{i,j,k} \binom{a}{i}\binom{b}{j}\binom{c}{k}$$

Since both methods count the exact same combinatorial set, the two expressions must be equal.

Problem 2: Pigeonhole Principle

We are given a subset $S \subseteq \{1, 2, 3, \ldots, 2n\}$ where $|S| = n + 1$. We want to show that there exist distinct $a, b \in S$ such that a/b is a power of 2.

Following the hint to factor each number into an odd number and a power of 2, any integer $x \in S$ can be uniquely expressed in the form $x = q \cdot 2^k$, where q is an odd integer and $k \geq 0$ is a non-negative integer.

We structure our Pigeonhole Principle argument as follows:

- **The Pigeons:** The $n + 1$ distinct elements of the set S.
- **The Holes:** The possible odd factors for numbers in the range 1 to $2n$. These are the odd integers $1, 3, 5, \ldots, 2n - 1$. Notice that there are exactly n odd numbers in this range.
- **The Rule:** Assign each pigeon $x \in S$ to the hole corresponding to its odd factor q (i.e., if $x = q \cdot 2^k$, it goes into hole q).

Since we are placing $n + 1$ pigeons into n holes, by the Pigeonhole Principle, at least one hole must contain at least two pigeons.

This means there are two distinct elements $a, b \in S$ that share the exact same odd factor q. We can write them as:

$$a = q \cdot 2^i$$

$$b = q \cdot 2^j$$

Since a and b are distinct elements of S, $i \neq j$. Without loss of generality, assume $i > j$. Then, dividing a by b gives:

$$\frac{a}{b} = \frac{q \cdot 2^i}{q \cdot 2^j} = 2^{i-j}$$

Since $i > j$, the difference $i - j$ is a positive integer, meaning a/b is a power of 2. This completes the proof.

Problem 3: Dice Dice Dice Dice Dice

(a) Probability Space:

- **Outcomes:** The set of outcomes consists of all ordered 5-tuples $(d_1, d_2, d_3, d_4, d_5)$ representing the results of the 5 dice rolls, where each $d_i \in \{1, 2, 3, 4, 5, 6\}$.
- **Number of outcomes:** Because there are 5 independent dice and each has 6 possible faces, there are $6^5 = 7776$ possible outcomes.
- **Probability:** Assuming the standard six-sided dice are fair, each outcome is equally likely. Thus, the probability of any single specific outcome is uniform: $\frac{1}{6^5} = \frac{1}{7776}$.

(b) A_{29} is the event that the sum of the dice is at least 29. The maximum possible sum is $5 \times 6 = 30$. Thus, A_{29} includes all outcomes where the sum is exactly 29 or 30.

- Sum = 30 requires all sixes: $(6, 6, 6, 6, 6)$.
- Sum = 29 requires four sixes and one five. There are 5 such permutations based on which die is the 5.

Explicitly, $A_{29} = \{(6,6,6,6,6), (5,6,6,6,6), (6,5,6,6,6), (6,6,5,6,6), (6,6,6,5,6), (6,6,6,6,5)\}$. Since there are exactly 6 outcomes in A_{29}, we have:

$$\Pr[A_{29}] = \frac{6}{6^5} = \frac{1}{6^4} = \frac{1}{1296}$$

(c) A_{18} is the event that the sum of the dice is at least 18. Let $A_{<18}$ be its complement (the event that the sum is exactly 17 or less). We define a function $f : A_{18} \to A_{<18}$ mapping an outcome $(d_1, d_2, d_3, d_4, d_5)$ to $(7-d_1, 7-d_2, 7-d_3, 7-d_4, 7-d_5)$.

Proof it is a bijection: First, verify it maps into the target set. If the sum of the original dice is $S \geq 18$, the sum of the new mapped dice is $\sum_{i=1}^{5}(7-d_i) = 35 - S$. Since $S \geq 18$, the new sum $35 - S \leq 35 - 18 = 17$. Thus, the output is strictly in $A_{<18}$. Second, verify it is invertible. Applying the function twice yields $7-(7-d_i) = d_i$. Because it is its own inverse, it maps 1-to-1 and onto, making it a bijection.

Because a bijection exists between A_{18} and its complement, both sets must have the exact same number of elements. Thus, $|A_{18}| = \frac{1}{2}|S_{total}|$, which means:

$$\Pr[A_{18}] = \frac{1}{2}$$

(d) Let B_k be the event of exactly k sixes, and C_k be the event of at least k sixes. To find B_k, we choose which k of the 5 dice show a six ($\binom{5}{k}$ ways). The chosen k dice have a 1/6 chance of being six, and the remaining $5-k$ dice have a 5/6 chance of not being six. Thus:

$$\Pr[B_k] = \binom{5}{k}\left(\frac{1}{6}\right)^k\left(\frac{5}{6}\right)^{5-k} = \frac{\binom{5}{k}5^{5-k}}{6^5}$$

Since C_k is the cumulative probability of having k or more sixes:

$$\Pr[C_k] = \sum_{i=k}^{5}\Pr[B_i] = \sum_{i=k}^{5}\frac{\binom{5}{i}5^{5-i}}{6^5}$$

(e) We want $\Pr[C_3 \mid C_2]$. Using the definition of conditional probability and noting that $C_3 \subset C_2$ (if you have at least 3 sixes, you definitely have at least 2), $C_3 \cap C_2 = C_3$.

$$\Pr[C_3 \mid C_2] = \frac{\Pr[C_3 \cap C_2]}{\Pr[C_2]} = \frac{\Pr[C_3]}{\Pr[C_2]}$$

Using the verified values given in the problem:

$$\Pr[C_3 \mid C_2] = \frac{276/6^5}{1526/6^5} = \frac{276}{1526} = \frac{138}{763} \approx 0.1809$$

(f) Let D_k be the event that Ben rolls at least k dice showing the same value. We want $\Pr[D_3 \mid D_2]$. First, find the total size of D_3. There are 6 values a die can show. For each value, the number of ways to get at least 3 of them is equal to $|C_3| = 276$. Since it is impossible to have at least 3 of one value AND at least 3 of a different value at the same time (which would require 6 dice), these 6 events are disjoint. Thus, $|D_3| = 6 \times 276 = 1656$.

Next, find $|D_2|$. It is easier to find its complement, $D_{<2}$, which means no two dice show the same value (all 5 dice are distinct). $|D_{<2}| = 6 \times 5 \times 4 \times 3 \times 2 = 720$. Thus, $|D_2| = 6^5 - 720 = 7776 - 720 = 7056$.

Since $D_3 \subset D_2$:

$$\Pr[D_3 \mid D_2] = \frac{|D_3 \cap D_2|}{|D_2|} = \frac{|D_3|}{|D_2|} = \frac{1656}{7056} = \frac{23}{98} \approx 0.2347$$

(g) Ben is down to 3 dice. The new sample space size is $6^3 = 216$. The pairs of dice faces that sum exactly to 6 are $\{1,5\}$, $\{2,4\}$, and $\{3,3\}$. Let E_{15} be the event that the 3 rolls include a 1 and a 5. Let E_{24} be the event that the 3 rolls include a 2 and a 4. Let E_{33} be the event that the 3 rolls include at least two 3s. We seek the size of the union $|E_{15} \cup E_{24} \cup E_{33}|$.

Using Inclusion-Exclusion:

- $|E_{15}|$: We can have (1,5,X). There are $3! = 6$ permutations if the 3 dice are distinct, and 4 choices for X (not 1 or 5), giving $6 \times 4 = 24$. If X is 1 (e.g., 1,1,5), there are 3 permutations. If X is 5, there are 3 permutations. Total $|E_{15}| = 24 + 3 + 3 = 30$.
- $|E_{24}|$: By identical symmetry to $\{1,5\}$, $|E_{24}| = 30$.
- $|E_{33}|$: We can have (3,3,X) where X is not 3. There are $\binom{3}{2} = 3$ ways to place the 3s, and 5 choices for X. $3 \times 5 = 15$. Plus the single outcome of (3,3,3) makes 16. Total $|E_{33}| = 16$.

Next, look at intersections. Is it possible to satisfy two of these events simultaneously with only 3 dice? $E_{15} \cap E_{24}$ requires a 1, 5, 2, and 4 (4 dice). Thus, it's empty (size 0). $E_{15} \cap E_{33}$ requires a 1, 5, 3, 3. Empty (size 0). $E_{24} \cap E_{33}$ requires 2, 4, 3, 3. Empty (size 0).

The total number of valid outcomes is simply $30 + 30 + 16 = 76$.

$$\Pr[\text{pair sums to 6}] = \frac{76}{216} = \frac{19}{54}$$

Problem 4: Unreliable Helpers

(a) We translate the problem statement into formal probability equations. Given $\Pr[E] = 0.10$ and $\Pr[\bar{E}] = 0.90$.

The TeaBot malfunctions 20% of the time. This means it gives the opposite answer of reality:

$$\Pr[T \mid \bar{E}] = 0.20 \quad \text{(Says error when there isn't one)}$$

$$\Pr[\bar{T} \mid E] = 0.20 \implies \Pr[T \mid E] = 1 - 0.20 = 0.80 \quad \text{(Says error when there is one)}$$

The LecBot malfunctions 25% of the time:

$$\Pr[L \mid \bar{E}] = 0.25$$

$$\Pr[\bar{L} \mid E] = 0.25 \implies \Pr[L \mid E] = 1 - 0.25 = 0.75$$

(b) We want to find $\Pr[E \mid \bar{T} \cap L]$. Since the bots malfunction independently given the state of the problem, we can use conditional independence: $\Pr[\bar{T} \cap L \mid E] = \Pr[\bar{T} \mid E] \Pr[L \mid E] = (0.20)(0.75) = 0.15$ $\Pr[\bar{T} \cap L \mid \bar{E}] = \Pr[\bar{T} \mid \bar{E}] \Pr[L \mid \bar{E}] = (0.80)(0.25) = 0.20$

By Bayes' Theorem:

$$\Pr[E \mid \bar{T} \cap L] = \frac{\Pr[\bar{T} \cap L \mid E] \Pr[E]}{\Pr[\bar{T} \cap L \mid E] \Pr[E] + \Pr[\bar{T} \cap L \mid \bar{E}] \Pr[\bar{E}]}$$

$$\Pr[E \mid \bar{T} \cap L] = \frac{(0.15)(0.10)}{(0.15)(0.10) + (0.20)(0.90)}$$

$$\Pr[E \mid \bar{T} \cap L] = \frac{0.015}{0.015 + 0.180} = \frac{0.015}{0.195} = \frac{15}{195} = \frac{1}{13}$$

(c) Is $\Pr[T \mid L]$ equal to $\Pr[T]$?

Intuition: No, they are not equal. While the robots malfunction independently, they are both evaluating the exact same underlying problem. If the LecBot says there is an error (L), this is evidence that the problem likely does indeed have an error (E). Because the problem is now more likely to have an error, the TeaBot (who is mostly accurate) is correspondingly more likely to also report an error. Thus, observing L provides information about E, which in turn provides information about T, breaking unconditional independence. We expect $\Pr[T \mid L] > \Pr[T]$.

Numerical Verification: First, find the unconditional probability $\Pr[T]$ using the Law of Total Probability:

$$\Pr[T] = \Pr[T \mid E] \Pr[E] + \Pr[T \mid \bar{E}] \Pr[\bar{E}]$$

$$\Pr[T] = (0.80)(0.10) + (0.20)(0.90) = 0.08 + 0.18 = 0.26$$

Next, we find $\Pr[L]$:

$$\Pr[L] = \Pr[L \mid E] \Pr[E] + \Pr[L \mid \bar{E}] \Pr[\bar{E}]$$

$$\Pr[L] = (0.75)(0.10) + (0.25)(0.90) = 0.075 + 0.225 = 0.30$$

Next, we find the intersection $\Pr[T \cap L]$ using conditional independence:

$$\Pr[T \cap L] = \Pr[T \cap L \mid E] \Pr[E] + \Pr[T \cap L \mid \bar{E}] \Pr[\bar{E}]$$

$$\Pr[T \cap L] = (\Pr[T \mid E] \Pr[L \mid E]) \Pr[E] + (\Pr[T \mid \bar{E}] \Pr[L \mid \bar{E}]) \Pr[\bar{E}]$$

$$\Pr[T \cap L] = (0.80 \times 0.75)(0.10) + (0.20 \times 0.25)(0.90)$$

$$\Pr[T \cap L] = (0.60)(0.10) + (0.05)(0.90) = 0.06 + 0.045 = 0.105$$

Now we calculate $\Pr[T \mid L]$:

$$\Pr[T \mid L] = \frac{\Pr[T \cap L]}{\Pr[L]} = \frac{0.105}{0.30} = 0.35$$

Since $0.35 \neq 0.26$, we mathematically verify that $\Pr[T \mid L] \neq \Pr[T]$.